MSG No. 129.415 $P4'/nmm'$ [Type III, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `-4'm2'`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[1,4,6,7,11,12,13,14]`
2	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[2,3,5,8,9,10,15,16]`

Table 2: Wyckoff site: `2b`, site symmetry: `-4'm2'`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[1,4,6,7,11,12,13,14]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[2,3,5,8,9,10,15,16]`

Table 3: Wyckoff site: `2c`, site symmetry: `4'mm'`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	`[1,4,6,7,9,10,15,16]`
2	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	`[2,3,5,8,11,12,13,14]`

Table 4: Wyckoff site: `4d`, site symmetry: `..2'/m'`

No.	position	mapping
1	$[0, 0, 0]$	`[1,5,12,16]`
2	$\left[\frac{1}{2}, 0, 0\right]$	`[2,6,9,13]`
3	$\left[0, \frac{1}{2}, 0\right]$	`[3,7,10,14]`
4	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[4,8,11,15]`

Table 5: Wyckoff site: `4e`, site symmetry: `..2'/m'`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,5,12,16]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[2,6,9,13]`
3	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[3,7,10,14]`
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[4,8,11,15]`

Table 6: Wyckoff site: `4f`, site symmetry: `2mm.`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	`[1,4,6,7]`
2	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	`[2,3,5,8]`
3	$\left[\frac{1}{4}, \frac{3}{4}, z\right]$	`[9,10,15,16]`
4	$\left[\frac{3}{4}, \frac{1}{4}, -z\right]$	`[11,12,13,14]`

Table 7: Wyckoff site: `8g`, site symmetry: `..2'`

No.	position	mapping
1	$[x,\ -x,\ 0]$	`[1,12]`
2	$\left[x+\frac{1}{2},\ x,\ 0\right]$	`[2,9]`
3	$\left[-x,\ \frac{1}{2}-x,\ 0\right]$	`[3,10]`
4	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ 0\right]$	`[4,11]`
5	$[-x,\ x,\ 0]$	`[5,16]`
6	$\left[\frac{1}{2}-x,\ -x,\ 0\right]$	`[6,13]`
7	$\left[x,\ x+\frac{1}{2},\ 0\right]$	`[7,14]`
8	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$	`[8,15]`

Table 8: Wyckoff site: `8h`, site symmetry: `..2'`

No.	position	mapping
1	$\left[x,\ -x,\ \frac{1}{2}\right]$	`[1,12]`
2	$\left[x+\frac{1}{2},\ x,\ \frac{1}{2}\right]$	`[2,9]`
3	$\left[-x,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[3,10]`
4	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[4,11]`
5	$\left[-x,\ x,\ \frac{1}{2}\right]$	`[5,16]`
6	$\left[\frac{1}{2}-x,\ -x,\ \frac{1}{2}\right]$	`[6,13]`
7	$\left[x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[7,14]`
8	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[8,15]`

Table 9: Wyckoff site: `8i`, site symmetry: `.m.`

No.	position	mapping
1	$\left[\frac{1}{4},\ y,\ z\right]$	`[1,6]`
2	$\left[\frac{3}{4},\ -y,\ -z\right]$	`[2,5]`
3	$\left[\frac{3}{4},\ y+\frac{1}{2},\ -z\right]$	`[3,8]`
4	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ z\right]$	`[4,7]`
5	$\left[\frac{1}{2}-y,\ \frac{1}{4},\ z\right]$	`[9,15]`
6	$\left[y,\ \frac{1}{4},\ z\right]$	`[10,16]`
7	$\left[y+\frac{1}{2},\ \frac{3}{4},\ -z\right]$	`[11,13]`
8	$\left[-y,\ \frac{3}{4},\ -z\right]$	`[12,14]`

Table 10: Wyckoff site: `8j`, site symmetry: `..m'`

No.	position	mapping
1	$[x,\ x,\ z]$	`[1,16]`
2	$\left[x+\frac{1}{2},\ -x,\ -z\right]$	`[2,13]`
3	$\left[-x,\ x+\frac{1}{2},\ -z\right]$	`[3,14]`
4	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ z\right]$	`[4,15]`
5	$[-x,\ -x,\ -z]$	`[5,12]`
6	$\left[\frac{1}{2}-x,\ x,\ z\right]$	`[6,9]`
7	$\left[x,\ \frac{1}{2}-x,\ z\right]$	`[7,10]`

continued ...

Table 10

No.	position	mapping
8	$\left[x+\frac{1}{2}, x+\frac{1}{2}, -z\right]$	`[8,11]`

Table 11: Wyckoff site: `16k`, site symmetry: `1`

No.	position	mapping
1	$[x, y, z]$	`[1]`
2	$\left[x+\frac{1}{2}, -y, -z\right]$	`[2]`
3	$\left[-x, y+\frac{1}{2}, -z\right]$	`[3]`
4	$\left[\frac{1}{2}-x, \frac{1}{2}-y, z\right]$	`[4]`
5	$[-x, -y, -z]$	`[5]`
6	$\left[\frac{1}{2}-x, y, z\right]$	`[6]`
7	$\left[x, \frac{1}{2}-y, z\right]$	`[7]`
8	$\left[x+\frac{1}{2}, y+\frac{1}{2}, -z\right]$	`[8]`
9	$\left[\frac{1}{2}-y, x, z\right]$	`[9]`
10	$\left[y, \frac{1}{2}-x, z\right]$	`[10]`
11	$\left[y+\frac{1}{2}, x+\frac{1}{2}, -z\right]$	`[11]`
12	$[-y, -x, -z]$	`[12]`
13	$\left[y+\frac{1}{2}, -x, -z\right]$	`[13]`
14	$\left[-y, x+\frac{1}{2}, -z\right]$	`[14]`
15	$\left[\frac{1}{2}-y, \frac{1}{2}-x, z\right]$	`[15]`
16	$[y, x, z]$	`[16]`